

EE798I Project Abstract

Design and Co-Optimization of a Fabrication-Tolerant, High-Q SiN Microring Resonator for Purcell Enhancement of a Diamond NV Center

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Abstract

The nitrogen-vacancy (NV) center in diamond is a leading solid-state qubit for quantum information processing and sensing, yet its scalability is hindered by the low efficiency of its coherent optical transition, the zero-phonon line (ZPL). This work proposes the design and comprehensive co-optimization of a hybrid quantum photonic interface to address this challenge. The system comprises a high-quality-factor (Q) silicon nitride microring resonator evanescently coupled to an NV center hosted in a diamond nanocrystal. The primary objective is to maximize the Purcell enhancement of the NV center's ZPL emission at 637 nm by engineering a resonator with a high Q/V ratio, where V is the effective mode volume. We shall design and simulate a multi-physics and statistical co-optimization framework that moves beyond ideal device performance to ensure practical viability. The methodology integrates **(1)** 3D FDTD simulations to model the coupled light-matter interaction and calculate the Purcell factor, **(2)** a multiphysics model of an integrated micro-heater for thermo-optic tuning to compensate for fabrication deviations, and **(3)** a Monte Carlo-based fabrication tolerance analysis to produce a design that is robust against typical nanometer-scale manufacturing errors. The optimization targets a figure of merit that simultaneously rewards high mean performance and penalizes high statistical variance. The use of nitrogen-rich SiN is specified to mitigate material autofluorescence, ensuring a high signal-to-noise ratio. This work provides a complete design pathway for a manufacturable, high-performance spin-photon interface, representing a critical step towards scalable quantum technologies based on solid-state emitters.

References

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